Convex Optimization

{Ref: Appendix B of "ML Foundations"}

Def A set $\mathcal{X} \subseteq \mathbb{R}^n$ is convex if, for all $x, y \in \mathcal{X}$,

$$\alpha x + (1-\alpha) y \in \mathcal{X} \quad \forall \alpha \in [0,1].$$

In other words, a set is convex if the line segment joining any two points in the set lies within the set.

Eg: Let A be an $m \times n$ matrix, b an $m \times 1$ vector.

$$\{x \in \mathbb{R}^n : Ax + b \leq 0\} \text{ is convex}$$

pointwise

Note: The set is the intersection of half-spaces. Also called a polyhedral set.

Operations that preserve convexity

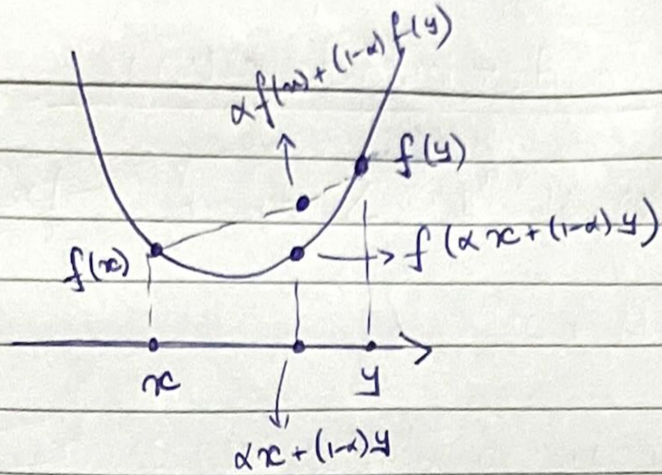
Lemma: Say C_1 & C_2 are convex sets. Then
- $C_1 \cap C_2$ is convex
- $C_1 \times C_2$ is convex
- $C_1 + C_2 := \{x_1 + x_2 : x_1 \in C_1, x_2 \in C_2\}$ is convex

↳ Exercise

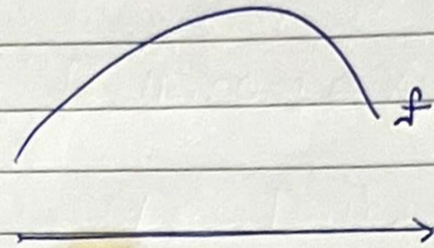
Def Say $\mathcal{X}$ is convex. A function $f: \mathcal{X} \to \mathbb{R}$ is convex if, for all $x, y \in \mathcal{X}$, $\alpha \in [0,1]$,

$$f(\alpha x + (1-\alpha) y) \leq \alpha f(x) + (1-\alpha) f(y). \quad -(1)$$

↳ Graph of function lies below any chord joining 2 points on the graph.



CONVEX FUNCTION



CONCAVE FUNCTION

Def:
- f is strictly convex if ① is strict whenever $x \neq y$, $\alpha \in (0,1)$.
- f is concave $\Leftrightarrow$ $-f$ is convex.
- f is strictly concave $\Leftrightarrow$ $-f$ is strictly convex.

Theorem: Say $f : X \to \mathbb{R}$ is differentiable & X is convex. Then f is convex iff

$$f(y) \geq \underbrace{f(x) + (y-x)^T \nabla f(x)}_{\text{tangent hyperplane at } x} \qquad \forall x, y \in X.$$

Moral: A convex function is bounded from below by its tangent hyperplanes.

Theorem Say $f : X \to \mathbb{R}$ is twice differentiable, and X is convex. Then f is convex (resp. strictly convex) iff $\nabla^2 f$ is positive semidefinite (resp. positive definite) $\forall x$.

Eg For a positive semidefinite matrix Σ, the function $x^T \Sigma x$ is convex over $\mathbb{R}^n$.

Eg Any norm $\|\cdot\|$ over a convex set $\mathcal{X}$ is convex, since by triangle inequality,

$$\|\alpha x + (1-\alpha) y\| \leq \|\alpha x\| + \|(1-\alpha) y\| = \alpha\|x\| + (1-\alpha)\|y\|.$$

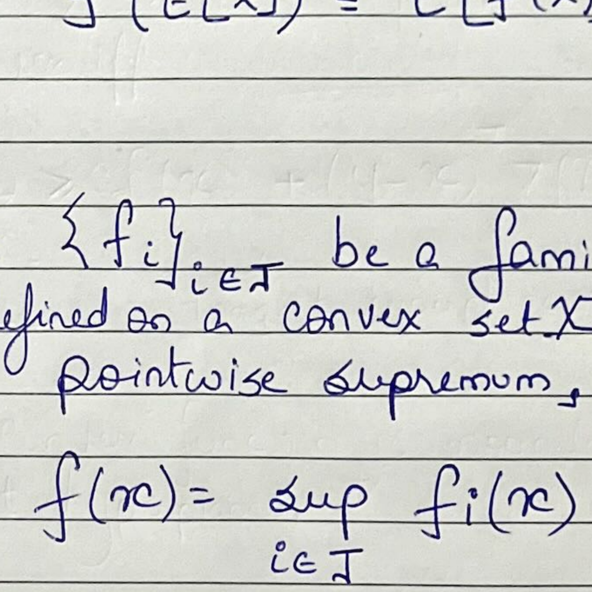
Theorem (Jensen's inequality) Let X be a random variable taking values in a convex set $\mathcal{X}$. Then for a convex function f over $\mathcal{X}$,

$$f(E[X]) \leq E[f(X)].$$

Lemma Let $\{f_i\}_{i \in I}$ be a family of convex functions defined on a convex set $\mathcal{X}$. Then the pointwise supremum, given by

$$f(x) = \sup_{i \in I} f_i(x)$$

is convex.



Moral: Pointwise supremum of convex functions is convex.

Now, suppose that $\mathcal{X}$ is convex; f is convex over $\mathcal{X}$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$

We consider the following convex optimization with linear constraints:

$$\min_{x \in \mathcal{X}} \underbrace{f(x)}_{\text{objective}} \qquad (P)$$

$$\text{subject to: } \left. Ax \leq b \right\} \text{ Constraints}$$

- This is called the primal problem (P)
- Linear constraint can be replaced by convex constraints $g_i(x) \leq 0$, $1 \leq i \leq m$, where $g_i(\cdot)$ are convex over $\mathcal{X}$.
  ↳ see handout
- Effectively (P) minimizes f over $\underbrace{\mathcal{X} \cap \{x : Ax \leq b\}}_{\text{Convex set}}$
- Let p^* denote the optimal value of (P).

Now, define the Lagrangian $\mathcal{L} : \mathcal{X} \times \mathbb{R}^m_+ \to \mathbb{R}$ as follows:

$$\mathcal{L}(x, \alpha) = f(x) + \sum_{1}^{m} \alpha_i (a_i x - b_i)$$

$$= f(x) + \alpha^T (Ax - b)$$

$$\left\{ A = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \right.$$

Note: Constraints have been replaced by "violation penalties"

Def: The (Lagrange) dual function $F: \mathbb{R}^m_+ \to \mathbb{R}$ is defined as:

$$F(\alpha) = \inf_{x \in X} \mathcal{L}(x, \alpha)$$

Note: • F is concave $\{$ pointwise inf. of linear functions

• $F(\alpha) \le p^*$ $\left\{ \begin{array}{l} \text{For feasible } x, \\ f(x) + \sum \alpha_i (a_i x - b_i) \le f(x) \end{array} \right.$

$\Rightarrow$ Inf of LHS over X $\le$ Inf of RHS over feasible points

The Dual optimization associated with (P) is:

$$\max_{\alpha \in \mathbb{R}^m_+} F(\alpha) \qquad (D)$$

• Let d^* denote optimal value of (D). We know

$$d^* \le p^*$$

• Turns out (so long as feasible set is non-empty) ~~so long as set of feasible~~ that

$$d^* = p^*$$ (strong duality; no duality gap)

Note: When dealing with convex constraints, we need a bit more for strong duality; see Slater's conditions.

Theorem x^* solves (P) if and only if $\exists\ \alpha^* \geq 0$ s.t.

KKT Conditions

$$\begin{cases} \nabla_x \mathcal{L}(x^*, \alpha^*) = \nabla_x f(x^*) + \sum \alpha_i^* a_i^T = 0 \\ \nabla_\alpha \mathcal{L}(x^*, \alpha^*) = Ax^* - b \leq 0 \\ \alpha^{*T}(Ax^* - b) = 0 \end{cases}$$

In this case, α^* solves (D) & (x^*, α^*) is a saddle point of $\mathcal{L}$.

Moral:
- Under mild conditions, solving (P) is equivalent to solving (D). Often, one may be more efficient / insightful to work with.
- Solving either is equivalent to finding (x^*, α^*) satisfying KKT conditions.